

**This assignment will not be graded and is only for practice.**

## 1. Key Points:

### Calculating minors and cofactors of a matrix:

If  $A$  is a square matrix, then the minor of the entry in the  $i$ -th row and  $j$ -th column (denoted by  $M_{ij}$ ) is the determinant of the submatrix formed by deleting the  $i$ -th row and  $j$ -th column.

The  $ij$ -th cofactor (denoted by  $C_{ij}$ ) is defined to be  $(-1)^{i+j}M_{ij}$ .

### Calculating the determinant:

$$\det(A) = \sum_{i=1}^n a_{ij}(-1)^{i+j}M_{ij} = \sum_{i=1}^n a_{ij}C_{ij}$$

Let us consider a  $3 \times 3$  matrix  $A$  as follows:

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

### Calculating $M_{11}$ and $C_{11}$ :

- Find out the sub-matrix by deleting the first row and the first column:

$$\begin{bmatrix} \cdot & \cdot & \cdot \\ \cdot & a_{22} & a_{23} \\ \cdot & a_{32} & a_{33} \end{bmatrix}$$

- Calculating the determinant of the sub-matrix:

$$\det \begin{bmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{bmatrix} = a_{22}a_{33} - a_{32}a_{23}$$

$$M_{11} = a_{22}a_{33} - a_{32}a_{23}$$

$$C_{11} = (-1)^{1+1}(a_{22}a_{33} - a_{32}a_{23}) = a_{22}a_{33} - a_{32}a_{23}$$

### Calculating $M_{23}$ and $C_{23}$ :

- Find out the sub-matrix by deleting the second row and the third column:

$$\begin{bmatrix} a_{11} & a_{12} & \cdot \\ \cdot & \cdot & \cdot \\ a_{31} & a_{32} & \cdot \end{bmatrix}$$

- Calculating the determinant of the sub-matrix:

$$\det \begin{bmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{bmatrix} = a_{11}a_{32} - a_{31}a_{12}$$

$$M_{23} = a_{11}a_{32} - a_{31}a_{12}$$

$$C_{23} = (-1)^{2+3}(a_{11}a_{32} - a_{31}a_{12}) = -a_{11}a_{32} + a_{31}a_{12}$$

### Calculating the determinant of $A$ :

Expanding with respect to the first row:

$$\det(A) = a_{11}(-1)^{1+1}M_{11} + a_{12}(-1)^{1+2}M_{12} + a_{13}(-1)^{1+3}M_{13}$$

Expanding with respect to the second row:

$$\det(A) = a_{21}(-1)^{2+1}M_{21} + a_{22}(-1)^{2+2}M_{22} + a_{23}(-1)^{2+3}M_{23}$$

Expanding with respect to the third row:

$$\det(A) = a_{31}(-1)^{3+1}M_{31} + a_{32}(-1)^{3+2}M_{32} + a_{33}(-1)^{3+3}M_{33}$$

The determinant can also be calculated by expanding along columns.

Expanding with respect to the first column:

$$\det(A) = a_{11}(-1)^{1+1}M_{11} + a_{21}(-1)^{2+1}M_{21} + a_{31}(-1)^{3+1}M_{31}$$

- 1) Consider a square matrix  $A = \begin{bmatrix} -1 & 2 & 0 \\ 2 & 0 & -1 \\ -1 & 0 & 1 \end{bmatrix}$ . Find out  $M_{23}$ .

[Hint: Find out the sub-matrix by deleting the second row and the third column:  $\begin{bmatrix} -1 & 2 & \cdot \\ \cdot & \cdot & \cdot \\ -1 & 0 & \cdot \end{bmatrix}$

Calculate the determinant of the sub-matrix. ]

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

**1 point**

- 2) Consider a square matrix  $A = \begin{bmatrix} -1 & 2 & 0 \\ 2 & 0 & -1 \\ -1 & 0 & 1 \end{bmatrix}$ . Find out  $C_{32}$ .

[Hint: Find out the sub-matrix by deleting the third row and the second column:

$\begin{bmatrix} -1 & \cdot & 0 \\ 2 & \cdot & -1 \\ \cdot & \cdot & \cdot \end{bmatrix}$  Calculate the determinant of the sub-matrix to find  $M_{32}$ . To find the cofactor

$C_{23}$  the sign has to be taken care of as follows:  $C_{32} = (-1)^{3+2}M_{32}$  ]

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) -1

**1 point**

- 3) Consider a square matrix  $A = \begin{bmatrix} -1 & 2 & 0 \\ 2 & 0 & -1 \\ -1 & 0 & 1 \end{bmatrix}$ . Find out the determinant of  $A$ .

[Hint: Expanding with respect to the first row:

$$\det(A) = a_{11}(-1)^{1+1}M_{11} + a_{12}(-1)^{1+2}M_{12} + a_{13}(-1)^{1+3}M_{13}$$

Expanding with respect to any row will also give you the answer]

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) -2

1 point

4) Consider the following square matrices:

1 point

$$A = \begin{bmatrix} 2013 & 2014 & 2015 \\ 2016 & 2017 & 2022 \\ 2019 & 2020 & 2021 \end{bmatrix}, B = \begin{bmatrix} 2016 & 2017 & 2022 \\ 2013 & 2014 & 2015 \\ 2019 & 2020 & 2021 \end{bmatrix} \text{ and } C = \begin{bmatrix} 4032 & 4034 & 4044 \\ 2013 & 2014 & 2015 \\ 2019 & 2020 & 2021 \end{bmatrix}$$

Choose the set of correct options.

Hint:

$B$  can be obtained by interchanging the first two rows.

$C$  can be obtained by multiplying 2 to the first row of  $B$ .

- ☐  $\det(B) = \det(A)$ .
- ☐  $\det(A) = -\det(B)$ .
- ☐  $\det(C) = -2\det(B)$ .
- ☐  $\det(C) = -2\det(A)$ .

No, the answer is incorrect.

Score: 0

Feedback:

- Sign of the determinant changes due to interchange of two rows.
- If a scalar  $c$  is multiplied with a row of a matrix, then the determinant of the new matrix will be  $c$  times the determinant of the earlier matrix.

Accepted Answers:

$$\det(A) = -\det(B).$$

$$\det(C) = -2\det(B).$$

$$\det(C) = -2\det(A)$$

## 2. Key Points:

**Cramers' Rule:** (This rule is applied to any system of linear equations with  $n$  equations and  $n$  variables. Here we recall the method for a system of linear equations with 3 equations and 3 variables.)

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1$$

Consider a system of linear equations as follows:  $a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2$

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3$$

Let the matrix representation of the above system be  $Ax = b$ , where  $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$ ,  $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ , and  $b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$ .

Let  $A_{x_i}$  be the matrix obtained by replacing the  $i$ -th column of  $A$  (i.e.,  $\begin{bmatrix} a_{1i} \\ a_{2i} \\ a_{3i} \end{bmatrix}$ ) by  $b$ , for  $i = 1, 2, 3$ .

$$A_{x_1} = \begin{bmatrix} b_1 & a_{12} & a_{13} \\ b_2 & a_{22} & a_{23} \\ b_3 & a_{32} & a_{33} \end{bmatrix}$$

$$A_{x_2} = \begin{bmatrix} a_{11} & b_1 & a_{13} \\ a_{21} & b_2 & a_{23} \\ a_{31} & b_3 & a_{33} \end{bmatrix}$$

$$A_{x_3} = \begin{bmatrix} a_{11} & a_{12} & b_1 \\ a_{21} & a_{22} & b_2 \\ a_{31} & a_{32} & b_3 \end{bmatrix}$$

If  $\det(A) \neq 0$ , then the solutions to the above system are  $x_i = \frac{\det A_{x_i}}{\det A}$ , for  $i = 1, 2, 3$ . i.e.,  $x_1 = \frac{\det A_{x_1}}{\det A}$ ,  $x_2 = \frac{\det A_{x_2}}{\det A}$ ,  $x_3 = \frac{\det A_{x_3}}{\det A}$

$$x_1 + x_3 = 1$$

Consider a system of linear equations  $-x_1 + x_2 - x_3 = 1$

$$-x_2 + x_3 = 1$$

Let matrix representation of the above system be  $Ax = b$ , where  $A = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix}$ ,  $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

and  $b = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ . Let  $A_{x_i}$  be the matrix obtained by replacing the  $i$ -th column of  $A$  (i.e.,  $\begin{bmatrix} a_{1i} \\ a_{2i} \\ a_{3i} \end{bmatrix}$ ) by  $b$ , for  $i = 1, 2, 3$ .

Use the above information to answer the following questions(5,6):

5) Choose the set of correct options

**1 point**

☐  $A_{x_1} = \begin{bmatrix} 1 & 1 & 0 \\ -1 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix}$

☐  $A_{x_1} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}$

☐  $A_{x_2} = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix}$

☐  $A_{x_3} = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & 1 \\ 0 & -1 & 1 \end{bmatrix}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$A_{x_1} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}$$

$$A_{x_3} = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & 1 \\ 0 & -1 & 1 \end{bmatrix}$$

6) Choose the set of correct options about the solutions to this system.

**1 point**

☐  $x_1 = -2.$

☐  $x_2 = -2.$

☐  $x_3 = 3.$

☐ None of the above.

No, the answer is incorrect.

Score: 0



**Accepted Answers:**

$$x_1 = -2.$$

$$x_3 = 3.$$

**3. Key Points:****Calculating the inverse of a matrix A:**

Find out the cofactor matrix  $C$ , whose  $ij$ -th element is  $C_{ij}$ : the  $ij$ -th cofactor of  $A$ .

Adjoint of  $A$  is transpose of  $C$ .

Calculate the determinant of  $A$  and check whether it is non-zero or not.

If determinant is non-zero then the inverse of  $A$  exists and is given by

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A).$$

7)

Which of the following is the cofactor matrix of  $\begin{bmatrix} 3 & 2 & 3 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{bmatrix}$ ?

**1 point**

[Hint: Recall the steps to calculate the cofactors, given in Key Points 1.]

☐  $\begin{bmatrix} 1 & 1 & 1 \\ -1 & -3 & -1 \\ -2 & 0 & 2 \end{bmatrix}$

☐  $\begin{bmatrix} -1 & 1 & 2 \\ 1 & -3 & 0 \\ 1 & 1 & -2 \end{bmatrix}$

☐  $\begin{bmatrix} -1 & 1 & 1 \\ 1 & -3 & 1 \\ 2 & 0 & -2 \end{bmatrix}$

☐  $\begin{bmatrix} 1 & -1 & -1 \\ -1 & 3 & -1 \\ -2 & 0 & 2 \end{bmatrix}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\begin{bmatrix} -1 & 1 & 1 \\ 1 & -3 & 1 \\ 2 & 0 & -2 \end{bmatrix}$$

8)

Which of the following is the adjoint matrix of  $\begin{bmatrix} 3 & 2 & 3 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{bmatrix}$ ?

**1 point**

[Hint: Find the transpose to the cofactor matrix.]

☐  $\begin{bmatrix} 1 & -1 & -2 \\ 1 & -3 & 0 \\ 1 & -1 & 2 \end{bmatrix}$

☐  $\begin{bmatrix} -1 & 1 & 1 \\ 1 & -3 & 1 \\ 2 & 0 & -2 \end{bmatrix}$

☐  $\begin{bmatrix} -1 & 1 & 2 \\ 1 & -3 & 0 \\ 1 & 1 & -2 \end{bmatrix}$

☐  $\begin{bmatrix} 1 & -1 & -2 \\ -1 & 3 & 0 \\ -1 & -1 & 2 \end{bmatrix}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\begin{bmatrix} -1 & 1 & 2 \\ 1 & -3 & 0 \\ 1 & 1 & -2 \end{bmatrix}$$

9) Find out the determinant of  $\begin{bmatrix} 3 & 2 & 3 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{bmatrix}$ .

[Hint: Recall the steps to calculate the determinant, given in Key Point 1.]

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

**1 point**

10) Which of the following is the inverse matrix of  $\begin{bmatrix} 3 & 2 & 3 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{bmatrix}$ ?

**1 point**

[Hint: If determinant is non-zero then the inverse of  $A$  exists and is given by  $A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$ .]

☐  $\frac{1}{2} \begin{bmatrix} 1 & -1 & -2 \\ 1 & -3 & 0 \\ 1 & -1 & 2 \end{bmatrix}$

☐

$$\frac{1}{2} \begin{bmatrix} -1 & 1 & 1 \\ 1 & -3 & 1 \\ 2 & 0 & -2 \end{bmatrix}$$

☐  $\frac{1}{2} \begin{bmatrix} -1 & 1 & 2 \\ 1 & -3 & 0 \\ 1 & 1 & -2 \end{bmatrix}$

☐  $\frac{1}{2} \begin{bmatrix} 1 & -1 & -2 \\ -1 & 3 & 0 \\ -1 & -1 & 2 \end{bmatrix}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\frac{1}{2} \begin{bmatrix} -1 & 1 & 2 \\ 1 & -3 & 0 \\ 1 & 1 & -2 \end{bmatrix}$$

11) Choose the set of correct options.

**1 point**

Hint:

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A).$$

$$\det(A^{-1}) = \frac{1}{\det(A)}$$

- ☐ If  $A = A^{-1}$ , then  $\det(A)$  must be 1.
- ☐ If  $A = A^{-1}$ , then  $A$  must be the identity matrix.
- ☐ If  $A = A^{-1}$ , then  $A^2 = I$ .
- ☐ If  $A^{-1} = \text{adj}(A)$ , then  $\det(A)$  must be 1.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If  $A = A^{-1}$ , then  $A^2 = I$ .

If  $A^{-1} = \text{adj}(A)$ , then  $\det(A)$  must be 1.

**Your score is: 0/11**